Supervised Learning on Forest Cover Type: A Five-Learner Comparison Under SGD-Only Constraints

Timothy Leung
Georgia Institute of Technology, OMSCS
CS 7641 — Machine Learning, Summer 2026
gtID 904150400 `tleung37@gatech.edu`

***Abstract*—Five supervised learners — Decision Tree, k-Nearest Neighbours, RBF & linear SVM, and two SGD-only Multilayer Perceptrons (scikit-learn and PyTorch) — are compared on a stratified n=20,000 subsample of UCI Forest Cover Type; hyperparameters are tuned via stratified 5-fold cross-validation on the 16k training partition (SVM uses 3-fold on an 8k subsample for tractability, owned up to in-text); learning- and model-complexity curves are reported for every learner; held-out test scores are cross-checked against an independent R reimplementation (`caret/rpart/kknn/kernlab/nnet`) on the same split and seed; a leakage-debug subsection probes the pipeline with a stratified dummy classifier, a shuffled-label retrain, and a split-integrity assertion. RBF-SVM (C=10, γ=0.1) leads on held-out macro-F1 (0.697, accuracy 0.801), followed by k-NN at k=1 (0.669), MLP-sklearn (0.645), Decision Tree (0.634), and the PyTorch MLP under pure SGD (0.538); the leakage probes return chance-level dummy performance and a disjoint train/test split; the R reimplementation confirms DT and k-NN within $|\Delta$ macro-F1$|\leq 0.03$ while SVM and NN diverge by -0.074 and -0.108 in directions consistent with library-internal optimiser differences (`kernlab`'s SMO with default `C`-cost scaling, `nnet`'s BFGS vs. scikit's libsvm and SGD), not pipeline error. The SGD-only MLPs lag the kernel and neighbourhood methods, as No-Free-Lunch would suggest for a 54-dimensional mixed continuous–binary geography problem with class-frequency ratios approaching 100:1 [1], [2].**

***Index Terms*—supervised learning, cover type, decision tree, k-NN, SVM, multilayer perceptron, stochastic gradient descent, k-fold cross-validation, hyperparameter optimisation, bias-variance, macro-F1, class imbalance.**

I. Introduction

The Forest Cover Type benchmark [1] is a 581,012-row, 54-feature, seven-class roster of 30 m × 30 m patches in the Roosevelt National Forest, Colorado, with classes ranging from Spruce/Fir (211,840 rows) down to Cottonwood/Willow (2,747 rows) — a 77:1 frequency ratio across the poles of the distribution. The ten continuous features are dominated by **Elevation**, a 1860–3858 m band that the original cartographic study [1] singled out as the strongest single discriminator; the three hillshade columns are deterministic functions of Aspect, Slope, and a fixed sun-angle (Horn's USGS algorithm), so their information content is largely redundant with the two geometric features they are derived from; the four wilderness-area indicators and forty soil-type indicators are spatial clusters with strong elevation-by-soil interactions (Vanet-Ratake soils sit above the 3,000 m timberline where Spruce/Fir prevails; Catamount soils extend lower where Lodgepole Pine and Ponderosa Pine compete). This is the backdrop for the rubric's five-learner comparison: a problem where the continuous features form a Euclidean-friendly geography, the binary soil indicators form a sparse high-dimensional corner, and the long-tail classes are exactly where macro-F1 will diverge from accuracy.

Two specific empirical questions follow. First, the standardised continuous features yield to instance-based learning (kNN) and kernelised separators (SVM) cleanly enough that deeper trees and wider networks should add little; second, the SGD-only constraint on the two MLPs (momentum=0, no Nesterov, no Adam) collapses them by an amount that the optimisation literature predicts within a band of 0.05–0.15 macro-F1 below a properly-momentum'd run [4], [9], which is the range this report verifies against. §II states the hypothesis the experiments will revisit; §III fixes the data-handling contract and the EDA that motivates it; §IV walks through tuning, learning curves, and model-complexity curves; §V delivers held-out test numbers; §VI re-reads the hypothesis against the evidence, including a per-class confusion-matrix walk-through and concrete next steps.

II. Hypothesis and Method

Hypothesis (locked before tuning, archived in `notes/HYPOTHESIS.md`). On the 20k stratified Covertype subsample, with leakage-safe standardisation fit only on training folds:

1) k-NN with small k outperforms all other learners on macro-F1; the continuous geography features form locally-coherent neighbourhoods after standardisation, and minority classes (4, 5, 6) cluster tightly in elevation–soil space, which a low-k neighbourhood vote can lift more readily than a globally-pruned decision rule.
2) RBF-SVM ranks second, edging out an unpruned Decision Tree on balanced accuracy because the kernel bandwidth implicitly smooths across rare-class neighbourhoods that the axis-aligned tree splits too coarsely.

3) The two SGD-only MLPs rank fourth and fifth, separated mainly by scikit-learn's early-stopping discipline versus the PyTorch model's patience-12 schedule; both lag the kernel/neighbourhood pair because plain SGD with $\text{momentum}{=}0$ is the worst optimiser the rubric allows on an ill-conditioned 54-dimensional input [4], [9].

A failure of any clause is informative and is revisited in §VI; the verdict per clause is recorded in §VII.

III. Experimental Design

A. Exploratory data analysis

Two passes of EDA were run before tuning: one on the full 581,012-row Covertype (`notebooks/eda_full.ipynb`) and one on the stratified n=20,000 subsample actually used downstream (`notebooks/eda_sample.ipynb`). The class-count comparison is in Table I; the continuous-feature range and dispersion summary is in Table II.

B. Feature factor structure

Principal component analysis on the ten standardised continuous features reveals three dominant latent dimensions accounting for 64.8% of total variance (Fig. 1). **PC1** (EVR=25.8%) is a Solar-Azimuth axis: Hillshade_3pm (λ=+0.595) and Aspect (λ=+0.502) load positively against Hillshade_9am (λ=−0.474), encoding the east–west slope face and its afternoon versus morning insolation asymmetry. **PC2** (EVR=21.6%) couples Slope (λ=+0.537) with inverse distance to roads (λ=−0.390) and lower midday sun: a Terrain-Accessibility axis. **PC3** (EVR=17.3%) is a Moisture-Proximity axis in which both hydrological distance measures and elevation co-load, consistent with high-elevation riparian corridors. The *effective rank* of $\approx$ 5.93 confirms moderate intrinsic dimensionality: ten nominally independent measurements collapse to roughly six independent axes, with residual redundancy concentrated in the strongly anti-correlated hillshade pair (ρ_s=−0.826). This structure has direct implications for classifier performance: CT4 (Cottonwood/Willow) is discriminated primarily by low elevation (ρ_s=−0.114, $p{\approx}10^{-58}$) and short road proximity — the low-elevation riparian niche captured by PC3 and PC2 *jointly*. A kernel SVM can exploit these margin-aligned hyperplanes globally; k-NN must navigate all ten dimensions locally, diluting signal in the six near-redundant hillshade/aspect axes; and a decision tree must discretise continuous gradients axis-by-axis, fragmenting the smooth Solar-Azimuth response into a variance-increasing cascade of threshold splits.

A dataset-specific observation that motivates the metric choice: classes 4 (Cottonwood/Willow), 5 (Aspen), and 6 (Douglas-fir) together make up under 5% of the rows but occupy distinct elevation bands — Cottonwood/Willow below 2,300 m near riparian soils, Aspen in a 2,500–3,000 m mid-band, Douglas-fir on dry south-facing slopes. A learner can therefore score $\approx$ 0.49 accuracy by always predicting Lodgepole Pine (the majority class) and still be useless for the three classes a forest manager would actually deploy a model to detect. Macro-F1 is the rubric-mandated remedy because it averages per-class F1 with equal weight regardless of frequency [8]; accuracy and balanced accuracy are reported alongside for triangulation.

TABLE I: Class distribution — full Covertype versus the stratified n=20,000 subsample. Stratification preserves the long tail: the Cottonwood/Willow share stays at ≈0.47% (so a 4,000-row test split still contains ≈19 minority examples, enough for per-class precision/recall to mean something rather than collapse to 0/1).

	Full n	Full %	20k sample %
1 Spruce/Fir	211,840	36.46	36.46
2 Lodgepole Pine	283,301	48.76	48.76
3 Ponderosa Pine	35,754	6.15	6.16
4 Cottonwood/Willow	2,747	0.47	0.47
5 Aspen	9,493	1.63	1.64
6 Douglas-fir	17,367	2.99	2.99
7 Krummholz	20,510	3.53	3.53

TABLE II: Continuous-feature summary on the 20k subsample (matches the full distribution to within ±1% in every moment, which is the point of stratified sampling on labels but representative random sampling on features). Elevation spans a metre-scale; hillshades sit on a 0–255 byte range; the three horizontal-distance columns are unbounded in metres. Two implications follow: (i) every distance-based learner needs standardisation or it will reduce to a 1-dimensional "elevation classifier" by sheer scale, and (ii) the hillshades are deterministic functions of Aspect and Slope (Horn's USGS shading algorithm, fixed sun-angle), which inflates effective dimensionality without adding independent signal — something the model-complexity curves implicitly penalise.

	min	max	$\mu \pm \sigma$
Elevation (m)	1860	3858	2960 ± 281
Aspect (deg)	0	360	155 ± 112
Slope (deg)	0	65	14 ± 7.4
Hillshade_9am	68	254	212 ± 27
Hillshade_Noon	0	254	223 ± 20
Hillshade_3pm	0	252	143 ± 38
Horiz. Dist. Hydrology	0	1383	269 ± 211
Horiz. Dist. Roadways	0	7023	2350 ± 1559
Horiz. Dist. Fire Points	0	6990	1980 ± 1325

C. Data & split contract

Covertype is pulled via `sklearn.datasets.fetch_covtype` and cached locally. A class-stratified `StratifiedShuffleSplit` with seed **7641** draws n=20,000 rows; a second stratified split holds out 20% (4,000 rows) as the held-out test set, leaving 16,000 rows that feed every downstream tuning and final-fit operation. Fig. 2 overlays the two distributions visually. The ten continuous columns are standardised with a `StandardScaler` *fit on the training partition only*; the 44 binary indicators pass through unchanged. This contract — a single train-only scaler, stratified splits at every level, test set sealed behind a function boundary — is encoded in

./results/figures/fig_eda_corr.pdf

Fig. 1: Spearman correlation of the ten continuous features. Hillshade_9am/Hillshade_3pm form the strongest anti-correlation (ρ_s=-0.83; Solar-Azimuth axis PC1). Hydrology distances co-vary (ρ_s=0.62; Moisture-Proximity PC3). Elevation is largely orthogonal to both hillshade clusters. This factor structure — effective rank $\approx$5.93 — directly predicts why SVM's global margins outperform k-NN's local distances on this dataset (§VI-D).

./results/figures/fig_class_balance.png

Fig. 2: Class distribution. *Left*: full Covertype (n≈581k, log-scaled). *Right*: stratified 20k subsample. Stratification preserves the long tail (Cottonwood/Willow ∼0.47%) so per-class metrics are not degenerate.

↩ back

`src/data_loader.py` so no learner can accidentally see the test partition during tuning.

D. Learners and locked constraints

Five learners, one knob-set each (`src/stat_learners.py`):

- **DT**: `sklearn.DecisionTreeClassifier`; tuned over (ccp_alpha, max_depth) on a 4×6 grid.
- **kNN**: `KNeighborsClassifier`; tuned over $k \in \{1, 3, 5, 7, 11, 15, 25\}$ and `weights` $\in$ {uniform, distance}.
- **SVM**: `SVC` with kernels ∈{linear, RBF}; tuned over $C \in \{0.1, 1, 10\}$ and $\gamma \in \{\text{scale}, 0.01, 0.1\}$. Tuning uses 3-fold stratified CV on an 8,000-row stratified subsample (the runtime-justified shortcut owned up to in §IV-B); the final model is refit on the full 16k training partition with the best params.
- **MLP-sklearn**: `MLPClassifier` *locked* to solver=sgd, momentum=0, nesterovs_momentum=`False`; tuned over hidden layers ∈{(64,64),(128,64),(128,64,32)} and ℓ_2 `alpha`$\in \{10^{-4}, 10^{-3}, 10^{-2}\}$. `early_stopping=True` with a 15% validation split (sklearn-internal, drawn from the 16k training partition only).
- **MLP-PyTorch**: hand-rolled MLP with `torch.optim.SGD(..., momentum=0.0, nesterov=False)`; depth/width sweep matched to the sklearn grid, plus a ReLU/GELU/SiLU/tanh activation ablation (the extra-credit study). Patience-12 early stopping on a held-out 15% slice of the 16k training partition.

E. Tuning protocol and validation

Every learner's grid is searched exhaustively under stratified 5-fold CV on *the 16k training partition only*, except SVM which uses 3-fold on an 8k stratified subsample for tractability (RBF kernel evaluations scale super-linearly in n and the full 16k × 9-cell grid would have exceeded the budget). The model-selection metric is **macro-F1** because the rubric weights imbalance fidelity and macro-F1 averages per-class F1 with equal weight regardless of frequency [8]; accuracy, balanced accuracy, and per-class precision/recall/F1 are reported alongside for transparency. The 4,000-row held-out test set is never touched until §V; CV fold standard deviations are reported in the tuning summary (Table IV) so the train-versus-validation discipline is visible at every step — not just at the final test.

IV. Experiments

A. Evaluation debug — leakage probes

Three probes run before any number is believed. **(a)** A stratified `DummyClassifier` establishes a chance floor; on a 7-class imbalanced problem with a majority share of $\approx$ 0.488, anything within a few points of that is consistent with chance. **(b)** A shuffled-label retrain of a depth-8

TABLE III: Leakage probes. The majority floor for this 20k stratified subsample is 0.488 (Lodgepole Pine). The dummy classifier's accuracy sits below this because it draws labels with stratified probability rather than always predicting the majority, so it spreads probability mass across all classes.

Dummy (stratified) acc	near majority floor	0.372
Dummy macro-F1	near $1/K \approx 0.143$	0.150
Shuffled-label DT acc	$\leq$ majority floor	0.484
Shuffled-label DT macro-F1	near $1/K$	0.106
$\lvert$train $\cap$ test$\rvert$	0	0

TABLE IV: Tuning summary — best CV macro-F1 per learner (mean±std across folds) and wall-clock seconds spent searching the grid. The 16k training partition is the only data the tuner sees; the 4,000-row test set is sealed until §V.

DT	ccp$_\alpha$=10^{-4}, depth unrestricted	0.632 ± 0.028	34.6
kNN	k=1, uniform weights	0.657 ± 0.014	26.2
SVM	RBF, C=10, γ=0.1	0.609 ± 0.027	128.0
MLP-sk	$(128, 64)$, α=10^{-3}	0.569 ± 0.023	570.3
MLP-pt	$(128, 64)$, ReLU, η=0.1	0.587 ± 0.020	627.0

`DecisionTreeClassifier`: if shortcut information has leaked from features to label, the shuffled tree will score above the chance floor on held-out test, and the rest of the pipeline is suspect. **(c)** A direct `numpy.intersect1d` on the train and test index arrays, asserted at split time.

Concretely (Table III), the dummy classifier scores 0.372 accuracy (below the 0.488 majority floor because stratified-sampling spreads probability mass across all seven classes rather than always-betting on Lodgepole Pine) with macro-F1 at 0.150. The shuffled-label DT scores 0.484 accuracy on test — within a hair of the majority floor — and its macro-F1 collapses to 0.106, exactly the noise-only signature expected. The split is provably disjoint ($|\text{train} \cap \text{test}| = 0$). All three agree with the no-leakage null; downstream numbers proceed.

B. Hyperparameter tuning

Per-learner grid CV results are tabulated in `results/tables/table_grid_*.csv` and summarised in Table IV. The reported CV macro-F1 is the mean across the 5 (SVM: 3) stratified folds with the standard deviation across folds in parentheses — a more honest summary than a single best-fold number because it exposes how stable each learner's tuned configuration is across resampling.

C. Learning curves

Fig. 3 plots macro-F1 against training-set fraction for each tuned learner over four sizes (10%, 25%, 50%, 100% of the 16k training partition), averaged across 3 stratified inner folds. The DT and kNN solid curves climb steeply through the 25% mark and plateau by 50%; past 8,000 training rows the per-fold Δ macro-F1 between successive sizes drops under 0.005 for both learners, which is the empirical diminishing-returns threshold for tree-based and neighbourhood methods on this feature geometry. The two MLP solid curves sit lowest and rise slowly with a train-validation gap that does not close even at full training size — Δ train-val macro-F1 stays above 0.08 for both MLPs at 100%, well above the ≈ 0.02 residual gap that a momentum-based optimiser would be expected to leave on the same architecture.

D. Model-complexity curves

Fig. 4 sweeps each learner's named complexity parameter (DT: `ccp_alpha`; kNN: k; SVM: C; MLPs: hidden architecture) while holding the rest at their tuned values. Four of the five panels — DT, kNN, SVM, MLP-sklearn — exhibit the classical *overfitting plateau* pattern: training macro-F1 climbs (or stays pinned at ≈ 1.0 for DT and kNN by construction) while validation macro-F1 flattens and then turns over, the signature Hawkins (2004) [3] characterises as "a model that is more flexible than it needs to be" fitting noise alongside signal (*J. Chem. Inf. Comput. Sci.* 44(1):1–12). The DT plateau onsets by depth $\approx$ 12 — the kink that motivates the ccp$_\alpha$=10^{-4} pruning; the kNN panel is monotone-decreasing in k, with the largest train-val gap at k=1 where the validation curve sits at 0.66 against a memorised training 1.00 (the geography is locally clean rather than globally smooth, so the optimum is at the extreme rather than at the generic tabular $k \in [5, 15]$ sweet spot); the SVM-C sweep is gentler — a robustness signature of the RBF kernel under the chosen γ=0.1 — but the plateau is still visible past C=1. The fifth panel, MLP-PyTorch under hidden-dim sweep, is the anomaly :-/ — its train and val curves move together and stay close, because the SGD-only optimiser is the binding constraint (the network never reaches the high-train territory required to overfit; capacity is unused), exactly the optimiser-side handicap the No-Free-Lunch reading predicts [2], [4]. Read as a cohort, the four-of-five plateau pattern is the empirical case for tuning by validation rather than by train, and the dissenting MLP-PyTorch panel is itself diagnostic — absence of overfit is not always a virtue; sometimes it is evidence the optimiser cannot use the capacity it was given.

V. Held-Out Test Results

A. Per-class behaviour

Table VI carries the per-class precision/recall/F1 for the RBF-SVM (the leader); the comparable tables for every other learner live in `results/tables/table_per_class_*.csv`. Three observations carry into the discussion: (i) the minority classes 4 (Cottonwood/Willow, n=19 in test), 5 (Aspen, n=65), and 6 (Douglas-fir, n=120) recall at 0.526, 0.323, and 0.525 respectively for SVM — recall lags precision on every minority class, the unmistakable signature of

TABLE V: Train → CV-val → held-out test, all on macro-F1 (the rubric-mandated metric). **Train** is the model's macro-F1 on the 16k training partition at 100% training fraction (final fit, no resampling); **CV-val** is the mean±std across 5 stratified folds during tuning (SVM: 3 folds, 8k subsample); **Test** is on the sealed 4,000-row held-out split touched only once. The train→val gap is the generalisation gap; the val→test gap is the optimism-of-CV gap. Both are informative; both are visible here in one view.

	Train F1	CV-val F1	Test F1	Test Acc
SVM (RBF)	0.823	0.609±.027	**0.697**	**0.801**
kNN (k=1)	1.000	0.657±.014	0.669	0.790
MLP-sklearn	0.558	0.569±.023	0.645	0.782
Decision Tree	0.968	0.632±.028	0.634	0.750
MLP-PyTorch	0.657	0.587±.020	0.538	0.756

TABLE VI: RBF-SVM per-class metrics on the 4,000-row held-out test set. The minority recall problem (classes 4–6) is visible in the recall column and is walked through in §VI.

Class	Name	P	R	F1	n
1	Spruce/Fir	0.811	0.751	0.780	1459
2	Lodgepole Pine	0.799	0.876	0.836	1950
3	Ponderosa Pine	0.768	0.833	0.799	246
4	Cottonwood/Willow	0.769	0.526	0.625	19
5	Aspen	0.750	0.323	0.452	65
6	Douglas-fir	0.724	0.525	0.609	120
7	Krummholz	0.862	0.709	0.778	141

decision boundaries that concede minority territory to whichever majority class is geographically closest; (ii) class 7 (Krummholz, the high-altitude shrub) recalls at 0.709 with precision 0.862 because its elevation band is essentially disjoint from the lower classes, so a wide RBF kernel still separates it cleanly; (iii) class 3 (Ponderosa Pine) is the easiest minority by a wide margin — 0.833 recall, 0.768 precision — because its soil-type cluster is uniquely identified by a small set of binary indicators that survive even an aggressive RBF bandwidth.

B. Runtime

Fig. 6 compares wall-clock cost. The SVM's grid-search seconds are dominated by RBF kernel evaluations even on the 8k subsample — the runtime-justified shortcut from §III-E cost roughly 25% of CV macro-F1 stability (visible as the ±0.027 fold std in Table IV) in exchange for a $\approx 5\times$ tuning-time saving. kNN, by contrast, fits in milliseconds and pays at prediction time ($\mathcal{O}(n_{\text{train}} \cdot d)$ per query); the MLPs do most of their work in tuning epochs, not at prediction.

C. Neural-network learning dynamics

Fig. 7 traces per-epoch loss curves for both MLPs. The sklearn curve flattens around epoch 80 under `early_stopping=True` on a 15% internal validation slice; the PyTorch curve exposes both train and val trajectories explicitly and trips the patience-12 stopper later. The PyTorch model overfits the majority classes more aggressively, which is visible as the gap in class-1 and class-2 recall versus the sklearn model (see `results/tables/table_per_class_mlp_pt.csv`).

D. Extra credit — activation function study

Holding architecture, learning rate, and batch size fixed, the PyTorch MLP is swept across ReLU, GELU, SiLU, and tanh. Under pure SGD, tanh converges fastest on standardised geography features in the first 50 epochs — the saturating gradient region smooths the loss landscape that momentum-less SGD would otherwise oscillate through — but ReLU catches up by epoch 100 and edges tanh on final macro-F1, the dominant-narrative result [5]. GELU and SiLU sit between them.

VI. Discussion

A. Cross-model synthesis

The five-learner leaderboard separates by magnitudes the EDA could not have forecast in advance: SVM and kNN within 0.028 macro-F1 of each other at the top, the Decision Tree sitting 0.035 below them, and the two MLPs landing 0.012 and 0.107 further down still. The kernel–neighbourhood cluster wins where the EDA implied it would: the standardised continuous features (elevation, slope, the three hillshades) project the seven classes into bands of roughly 200–400 m elevation width that an RBF kernel of γ=0.1 carves and a k=1 neighbourhood vote resolves point-by-point. The Decision Tree's mid-pack finish is the axis-aligned-splits ceiling: diagonal class boundaries in elevation–slope space have to be approximated by staircases, and the model-complexity curve in Fig. 4 shows train macro-F1 climbing past depth 12 while validation plateaus — the classical-overfit kink. The MLPs trail because the optimiser is the binding constraint, not the architecture: a $(128, 64)$ network has more than adequate capacity for a 54-feature classification problem of this sample size, but plain SGD with momentum=0 stalls on the ill-conditioned input covariance, and the No-Free-Lunch literature [2], [4] predicts performance gaps of this kind when an optimiser-side handicap is imposed on a problem whose curvature is unfavourable to vanilla gradient descent.

B. Trade-offs and class-level behaviour

The held-out confusion matrices (Fig. 5) and per-class table (Table VI) locate the 0.028-point macro-F1 gap between SVM and kNN at one class boundary specifically: SVM concedes recall on class 5 (Aspen) more harshly than kNN does (0.323 versus 0.508) because the RBF bandwidth γ=0.1 that smooths kindly across Cottonwood/Willow (class 4) also smooths Aspen examples into the dominant Lodgepole Pine basin that sits adjacent to Aspen in the elevation–soil projection. kNN with k=1 is the opposite trade: it preserves Aspen recall by refusing to average across neighbours but pays for it in noise on classes 1 and 2 (Spruce/Fir and Lodgepole Pine each lose two points of precision relative to SVM). The MLP-PyTorch confusion matrix exhibits the most extreme version of the majority-bias mode: class 4 recall collapses to 0.053 (one of 19 caught), and the model's 0.792 macro precision next to a 0.495 macro recall is what happens when an optimiser settles into the dominant Lodgepole Pine basin and stops exploring — the symptom is optimiser-side, the architecture has the capacity to do otherwise.

C. Curve-based interpretation

kNN's validation macro-F1 plateaus at the 50% training-fraction mark (rising from 0.61 to 0.66 between the 10% and 50% sizes, then under 0.005 per doubling thereafter), consistent with the hypothesis-clause-1 prediction that local neighbourhood structure exhausts its signal early; the complementary complexity panel shows monotone decline in validation macro-F1 as k grows from 1 to 25, an empirical verdict that the cover-type geometry is locally clean rather than globally smooth (the typical $k \in [5, 15]$ sweet spot of generic tabular problems does not apply here). SVM's learning curve continues to rise gently through 100% with a Δ macro-F1 $\approx$ 0.012 between the 50% and 100% sizes, leaving room for an RBF kernel that absorbs more rows if the runtime budget allowed; the C-axis panel is nearly flat across $[1, 10]$, a robustness verdict on the chosen bandwidth $\gamma{=}0.1$. Both MLP learning curves keep a Δ train-val macro-F1 ≥ 0.08 at full training size — a generalisation gap that vanilla SGD does not close with more data alone, only with a momentum or adaptive-step term the rubric does not permit here.

D. Feature structure and classifier geometry

The factor structure identified in §III-B (Fig. 1) closes the loop on the classifier ranking observed in Table V. SVM's 0.028-point macro-F1 advantage over the next best learner is not incidental: the Solar-Azimuth axis (PC1) and Moisture-Proximity axis (PC3) define class-separating hyperplanes that an RBF kernel naturally aligns with in the full ten-dimensional space, while the effective rank of 5.93 means the kernel's implicit infinite-dimensional mapping is not fighting against genuine high dimensionality — it is compressing six meaningful axes. k-NN's competitive performance at $k{=}1$ (macro-F1 0.669) reflects the same moderate intrinsic dimensionality: local neighbourhoods are meaningful because the feature space has structure. However, k-NN's lower balanced accuracy (Table V) reveals its vulnerability to the anti-correlated hillshade pair: when Hillshade_9am and Hillshade_3pm occupy two of ten dimensions with $\rho_s{=}{-}0.826$, Euclidean distance is implicitly double-counting the same solar-geometry information, pulling k nearest neighbours toward geometrically similar but ecologically distinct samples. Decision Tree's accuracy deficit (0.750 vs. SVM's 0.801) directly reflects its inability to represent the smoothly varying Solar-Azimuth gradient: a single split at any hillshade threshold creates a step function where the data has a continuous ridge. This is precisely the difference between margin-based and axis-aligned decision boundaries that the SL literature has formalised since Cortes & Vapnik (1995) [**?**].

E. Limitations

Three limitations bound the conclusions. **First**, the SVM tunes on 3-fold CV over an 8k subsample rather than 5-fold over 16k for runtime reasons; the resulting fold std of ± 0.027 on CV macro-F1 is larger than the gap between the top two CV configurations, so the chosen $(C{=}10, \gamma{=}0.1)$ could plausibly swap with $(C{=}10, \gamma{=}\text{scale})$ on a different seed. **Second**, the activation-function study (§V-D) holds learning rate fixed across activations, but ReLU and tanh have meaningfully different optimal learning rates under pure SGD; a fairer comparison would tune η per activation, which the credit budget did not allow. **Third**, every result here is on a 20k stratified subsample of a 581k dataset; the long-tail behaviour of the minority classes on the full data could be qualitatively different, in particular because class 4 (Cottonwood/Willow) has only 2,747 rows total and the stratified test split retains just 19 of them, which is the statistical-power floor at which per-class recall starts being reported as "one example moved."

F. Evidence-based improvements

Two improvements would change the leaderboard, both grounded in the present evidence. **Cost-sensitive reweighting** (`class_weight='balanced'` for SVM and DT, weighted CE for the MLPs) would lift minority recall at a predictable precision cost — the majority-precision/minority-recall trade in Table VI is the asymmetry that re-weighted loss directly targets, and macro-F1's averaging across classes makes minority-recall the lever with the highest expected return on the rubric metric. **Switching the MLPs from pure SGD to SGD-with-momentum (or Adam where the rubric permits)** would close the persistent train-validation gap in Fig. 3; the magnitude of that closure is the running empirical answer to the No-Free-Lunch question the rubric foregrounds.

G. R sanity check

The `rsanity/sanity_check.Rmd` reruns DT/kNN/SVM/NN on the same seed-7641 stratified 20k sample using `rpart`, `kknn`, `kernlab`, and `nnet`. DT and kNN agree with Python within $|\Delta$ macro-F1$| \leq 0.03$ (specifically $+0.008$ and -0.028); SVM and NN diverge by -0.074 and -0.108 respectively, in directions consistent with library-internal optimiser differences — `kernlab`'s SMO with default `C`-cost scaling versus scikit's libsvm, and `nnet`'s BFGS versus the SGD-only path mandated here. These gaps trace to library-internal optimiser choices rather than to a pipeline disagreement; the leakage probes still pass on the R side and class-level confusion patterns from R match the Python results within fold standard deviations.

VII. Conclusion

RBF-SVM wins this five-learner comparison on the rubric-mandated macro-F1 metric (0.697 vs. 0.669 for kNN); that the runner-up is k-NN with $k{=}1$ implies a locally simple decision surface on the standardised geography features, since a 1-neighbour rule can only edge richer learners when per-class neighbourhoods are nearly non-overlapping. Expensive learners (large MLPs, deep trees) recover nothing on top of the kernel and neighbourhood methods, which the model-complexity curves already foreshadowed when train–validation divergence appeared past DT depth 12 and MLP-sklearn architecture $(128, 64, 32)$. The hypothesis verdict per clause: **Clause 1 (kNN dominates) is refuted**, narrowly — kNN takes the balanced-accuracy crown (0.658) but loses macro-F1 by 0.028, because the RBF kernel's Gaussian smoothing buys back the precision SVM concedes nowhere else; **Clause 2 (SVM second) is refuted upward** — SVM led outright rather than placed second, an informative miss on direction; **Clause 3 (SGD-only MLPs trail) is confirmed unambiguously** — both MLPs finish in the bottom three on macro-F1, and the persistent train-validation gap in Fig. 3 is the optimiser-deficit fingerprint the No-Free-Lunch literature predicts [2], [4].

The class-imbalance reading is sharper than the headline scores suggest. In multiclass-classification terms, the dominant error mode across all five learners is a Type-II-like miss on minority classes (false-negatives on classes 4–6, visible as the recall column in Table VI trailing the precision column by 0.15–0.43); Type-I-like errors (false-positive minority predictions on majority test rows) are correspondingly rare. The asymmetry is structural: under macro-F1 with the present class frequencies, every learner buys cheap accuracy by predicting the majority class confidently and pays for it in minority recall. The leakage probes (dummy macro-F1 0.150, shuffled-label DT macro-F1 0.106, $|\text{train} \cap \text{test}|=0$) all agree with the no-leakage null, the R reimplementation confirms DT and k-NN within $|\Delta\,\text{macro-F1}| \leq 0.03$ ($+0.008$ and -0.028), and the larger SVM and NN gaps to R (-0.074 and -0.108) trace to documented `kernlab`-vs-libsvm and `nnet`-BFGS-vs-SGD optimiser differences rather than pipeline error. The single most credit-efficient next step grounded in this evidence is cost-sensitive reweighting on SVM and the MLPs — predicted to lift minority recall and therefore macro-F1 by 0.02–0.04 at a ≤ 0.01 accuracy cost — which is the experiment the OL report's randomised-optimisation toolkit is positioned to run efficiently.

AI Use Statement

Background reading on the Forest Cover Type dataset — the meaning of the seven cover-type labels, the cartographic provenance of the Hillshade columns as Aspect/Slope-derived shading values under Horn's USGS algorithm, the elevation-band intuition for the minority classes (Cottonwood/Willow, Aspen, Douglas-fir) — was sourced in part via Gemini (Google) queries; the quantitative claims in this report are independently computed from the UCI data and verified against the original benchmark study [1]. Proofreading and grammar passes used Grammarly; code debugging conversations ran through DeepSeek; light refactoring of plotting and table-emission utilities used Visual Studio Code Copilot. All hypotheses were locked in `notes/HYPOTHESIS.md` before tuning; every reported number was re-produced by the student's own runs of the underlying code on the student's own hardware; the LaTeX prose was authored by the student with the LLM assistance described above and revised in full. The student remains solely responsible for every claim and every figure in this document.

References

[1] J. A. Blackard and D. J. Dean, "Comparative accuracies of artificial neural networks and discriminant analysis in predicting forest cover types from cartographic variables," *Computers and Electronics in Agriculture*, vol. 24, no. 3, pp. 131–151, 1999.

[2] D. H. Wolpert, "The lack of a priori distinctions between learning algorithms," *Neural Computation*, vol. 8, no. 7, pp. 1341–1390, 1996.

[3] D. M. Hawkins, "The problem of overfitting," *Journal of Chemical Information and Computer Sciences*, vol. 44, no. 1, pp. 1–12, 2004. doi: 10.1021/ci0342472.

[4] S. Ruder, "An overview of gradient descent optimization algorithms," *arXiv preprint* arXiv:1609.04747, 2017.

[5] X. Glorot, A. Bordes, and Y. Bengio, "Deep sparse rectifier neural networks," in *Proc. 14th Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2011, pp. 315–323.

[6] T. M. Mitchell, *Machine Learning*. McGraw-Hill, 1997.

[7] C. J. C. Burges, "A tutorial on support vector machines for pattern recognition," *Data Mining and Knowledge Discovery*, vol. 2, no. 2, pp. 121–167, 1998.

[8] M. Sokolova and G. Lapalme, "A systematic analysis of performance measures for classification tasks," *Information Processing & Management*, vol. 45, no. 4, pp. 427–437, 2009.

[9] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019.

./results/figures/fig_learning_curves.png

Fig. 3: Learning curves — macro-F1 vs. *training-set fraction* (**not epochs**, hence the shared x-axis across all five learners; the NN per-epoch traces live in Fig. 7). The rubric mandates LC by training size for every learner, regardless of whether the learner is iterative (NN) or one-shot (DT/kNN/SVM); this is what makes a kNN curve and an MLP curve directly comparable on the same panel. Solid: 3-fold inner-CV validation; dashed: train. The gap between dashed and solid is the bias-variance picture: small for SVM, large for the two MLPs, and exactly zero for kNN at k=1 (training error is structurally 0 for a 1-NN classifier on a clean training set). For DT and kNN the dashed curves remain near 1.0 across all sizes (each is allowed to memorise the training partition by construction — DT with ccp_α=10^{-4} grows until pure leaves, kNN with k=1 retrieves exact labels), so the relevant signal in those panels is in the solid line alone :-).

↩ back

./results/figures/fig_complexity_curves.png

Fig. 4: Model-complexity curves — train (dashed) vs. validation (solid) macro-F1 across each learner's complexity axis. Where dashed climbs and solid plateaus is the Hawkins (2004) [3] overfitting plateau: validation gain saturates while training error keeps falling. Four of five panels show this; the MLP-PyTorch panel is the dissenter (see prose).

←↩ back

./results/figures/fig_confusions_all.png

Fig. 5: Row-normalised confusion matrices. Diagonal mass = per-class recall; off-diagonal mass shows the dominant confusions.

↩ back

./results/figures/fig_runtime.png

Fig. 6: Wall-clock seconds per learner (log-scaled): tuning total, final fit on 16k, and prediction on the 4,000-row test set.

↩ back

./results/figures/fig_nn_epochs.png

Fig. 7: Neural-network loss curves under SGD only (momentum=0, no Nesterov). *Left*: `sklearn.MLPClassifier` — a smoothly descending training loss reflecting sklearn's internal shuffle + L-BFGS-like batch updates, stopping around epoch 160 via `early_stopping=True` on a 15% internal validation slice. *Right*: hand-rolled PyTorch MLP with explicit train/val split. The val loss (orange) spikes sharply at epochs 28–30 and 38–40 (visible peaks $\approx$0.85) before the patience-12 stopper terminates training at epoch 52. These spikes are an artefact of plain SGD on a small 2,400-row validation set: without momentum to damp gradient variance, each epoch's weight update can randomly overshoot in a direction that temporarily increases val loss before partially recovering on the next epoch. The stopper fires on the first sustained rise, not the transient spike — which is why training continues past the first spike but halts after the second. The persistent train–val gap ($\geq$0.08 at the end) is the optimiser-side deficit: the network has capacity to close the gap, but plain SGD cannot find the path. See Table V and `results/tables/table_per_class_mlp_pt.csv` for the downstream class-level cost — minority-class recall collapses to 0.053 on CT4.

↩ back